Master's Comprehensive Examination

Part I | Theory (90 minutes) | March 7, 1998

Instructions: Answer all four questions. Open book and open notes.

1. (a) Given $P(A) = .59$, $P(B) = .30$ and $P(A \cap B) = .21$, find (i) $P(A \cup B)$, (ii) $P(A \cap B')$.

(b) Draw two cards at random from a deck. What is the probability that both cards are strictly bigger than 3 and strictly less than 8?

2. Find the distribution function of a random variable X whose probability density is

$$f(x) = \begin{cases} \frac{1}{3}, & 0 < x < 1 \\ \frac{1}{3}, & 2 < x < 4 \\ 0, & \text{elsewhere.} \end{cases}$$

3. Let $X_1, \ldots, X_n$ constitute a random sample from a population with mean μ and a variance σ^2. (a) What condition on the constants $a_1, a_2, \ldots, a_n$ must be imposed so that

$$a_1 X_1 + a_2 X_2 + \cdots + a_n X_n$$

is an unbiased estimator of μ?

(b) Find the variance

$$X_1 + 2X_2 - 3X_3.$$

4. Let X be any observations on a Poisson random variable with parameter λ; i.e., X_i has density

$$f_X(x) = \frac{e^{-\lambda}\lambda^X}{x!}.$$

Use the Neyman-Pearson lemma to show that the best test of

$$H_0 : \lambda = 1 \text{ vs } H_1 : \lambda = 2,$$

is to reject when $X > C$.

If the size of the test is

$$1 - \left(e^{-1} + e^{-1} + \frac{e^{-1}}{2}\right),$$

what is C?

Master's Comprehensive Examination

Part II

Applied (90 minutes)

March 7, 1998

Instructions: Answer all four questions below. Open book and open notes.

1. Let P denote the probability that a die will come up 1. We seek to test the hypothesis:

$$H_0 : p = 1/6,$$
$$H_a : p > 1/6.$$

We will roll the die 100 times and observe x, the number of times a 1 comes up. The test will reject the null hypothesis if x exceeds the critical value c.

(a) Find c such that the test will have a 5% significance level.

(b) For the value of c found in part (a), find the power of the test against the alternative that $p = 1/4$.

(c) If $x = 25$, what would the observed level of significance (p-value) be?

2. It is known that birthdays are uniformly distributed throughout the year. A sample of 200 people is obtained and their birthdays recorded. The following summarizes these data:

Quarter	Count
Jan.-March	40
April-June	45
July-Sept.	40
Oct.-Dec.	75

It is claimed that this is a random sample. Test this claim.

3. An experiment was run to compare two drugs with respect to the reduction in cholesterol levels in the blood. The following are reductions over a 6-month period.

Drug A	Drug B
25	0
40	15
70	30
80	25
100	45
125	35
	30

Use an appropriate nonparametric procedure to compare the two drugs. Be sure to state the hypotheses that you are testing.

4. Two diets were compared with respect to milk production in cows. Ten cows were used in the experiment. Each cow used one diet during the first production period and the alternate diet during the second. The diet order was randomized. The data are pounds of milk.

Cow	Order	Diet 1	Diet 2
1	2	6	8
2	1	7	3
3	1	4	5
4	2	6	9
5	2	7	8
6	1	4	7
7	2	6	5
8	1	5	10
9	1	7	8
10	2	5	7

where order = 1 means diet 1 followed by diet 2; order = 2 means diet 2 followed by diet 1.

Construct the appropriate analysis of variance to determine if the mean response for the two diets is equal. Be sure to state all hypotheses tested and indicate the conclusions from your analysis.